

GEARS Citation Audit: Pipeline Bottlenecks and Rescue Procedures

Not every reduction represents a truly “lost” paper. The first transition mainly removes duplicate citation edges; later transitions reflect retrieval failures, machine-review failures, or evidence-based abstentions.

Stage-by-stage rescue summary

Stage	Reduction	Rescue procedures used	Why studies may still stop here
936 citation edges → 353 unique studies	583	Reconciled DOI, normalized title/authors/year, and linked preprints with published versions while preserving candidate URLs.	Mostly duplicate citation edges or paper versions. Genuine loss may result from missing DOI, incorrect metadata, or erroneous merging.
353 studies → 152 OA candidates	201	Expanded searches across OpenAlex OA locations, Europe PMC/PMC XML, publisher PDFs, preprint servers, and institutional repositories.	No indexed OA link, continuing paywall, missing public version, or mismatched DOI/title metadata.
152 OA candidates → 85 verified downloads	67	Skipped manual rights review under the documented override, tried alternative URLs and PDF/XML versions, and verified file integrity and paper identity.	Paywalls, HTTP errors, bot protection, expired links, corrupted files, HTML returned instead of PDF, wrong papers, supplements only, or unextractable scans.
85 machine-review entries → 55 machine-resolved	30	Retried failed agents, repaired JSON/schema outputs, re-extracted text, searched method aliases and citation contexts, and reran adjudication.	API failures, invalid outputs, poor OCR, excessive document length, missing citation context, or unresolved agent disagreement.
55 machine-resolved → 40 substantively classified	15	Searched additional versions and supplements. Allowed classification from incomplete coverage when available text directly demonstrated GEARS use; clear citation-only cases could be MENTION_ONLY.	Missing versions or supplements, citation not located, incorrect provider edge, ambiguous method naming, or contradictory evidence.

How scGPT and scVI differed

- scGPT used broad multi-source retrieval, alternative-version rescue, structured-output repair, and partial-coverage classification earlier.
- scVI initially used stricter retrieval and validation rules. We later applied the scGPT rescue strategy by merging first-pass and rescue documents, relaxing mention-only validation, rerunning failed agents, and classifying directly supported cases despite incomplete coverage.

Recommended GEARS rescue queues

Rescue queue	Number	Primary action
NO_OA_CANDIDATE	201	Search additional full-text sources and publication versions.
DOWNLOAD_OR_IDENTITY_FAILED	67	Retry downloads and resolve document identity or file-integrity failures.
MACHINE_UNRESOLVED	30	Retry agents, repair structured outputs, and readjudicate unresolved cases.
COVERAGE_INCOMPLETE / CITATION_NOT_LOCATED	15	Retrieve missing materials and repeat citation localization.

Interpretation

The final 40 studies represent 11.3% of the 353 unique citing studies. This set may be biased toward open-access, machine-readable papers with explicit and easily located descriptions of GEARS use. Reports should therefore present the 40 substantive classifications together with the remaining rescue and abstention queues.